

SciOdyssey.

Autonomous ML Research Agent

Preserve the research world.
Reset the researcher.

LONGER CONTEXTS. BRIGHTER SCIENCE.
SAME WORLD. MORE MINDS.

SAME
EVIDENCE

MORE
EXPLORERS
A BRIGHTER
TOMORROW

Science
Lives
Longer.

01 / PROBLEM

Different lifetimes.
Shared evidence.

Long-lived reasoning can anchor search. Tree search rereads files that cost more tokens to understand than edit. SciOdyssey retains the world and restarts with a fresh Scientist.



KNOWLEDGE
OUTLIVES
INDIVIDUALS.

02 / MOTIVATION

Research world

PERSISTENT
ON DISK

EVIDENCE	Knowledge + experiment ledger Metrics, failures and original reports
CURATED	Research brief + working priors Current understanding, open to revision
CODE	Retained implementation + State Recoverable version, evidence and provenance

Context reset — inherit experience, not prior reasoning.

03 / SYSTEM

Scientist

SCIENTIFIC JUDGMENT

- Read the research world
- Form hypotheses
- Code, train and evaluate
- Interpret and report evidence

Chooses the science within one trajectory.

META

PERSISTENCE AUTHORITY

- Audit claims against evidence
- Scope conclusions
- Curate shared memory
- Preserve optional State

Maintains what survives across trajectories.

Audit → scope → preserve → fresh Scientist

04 / EXPERIENCE

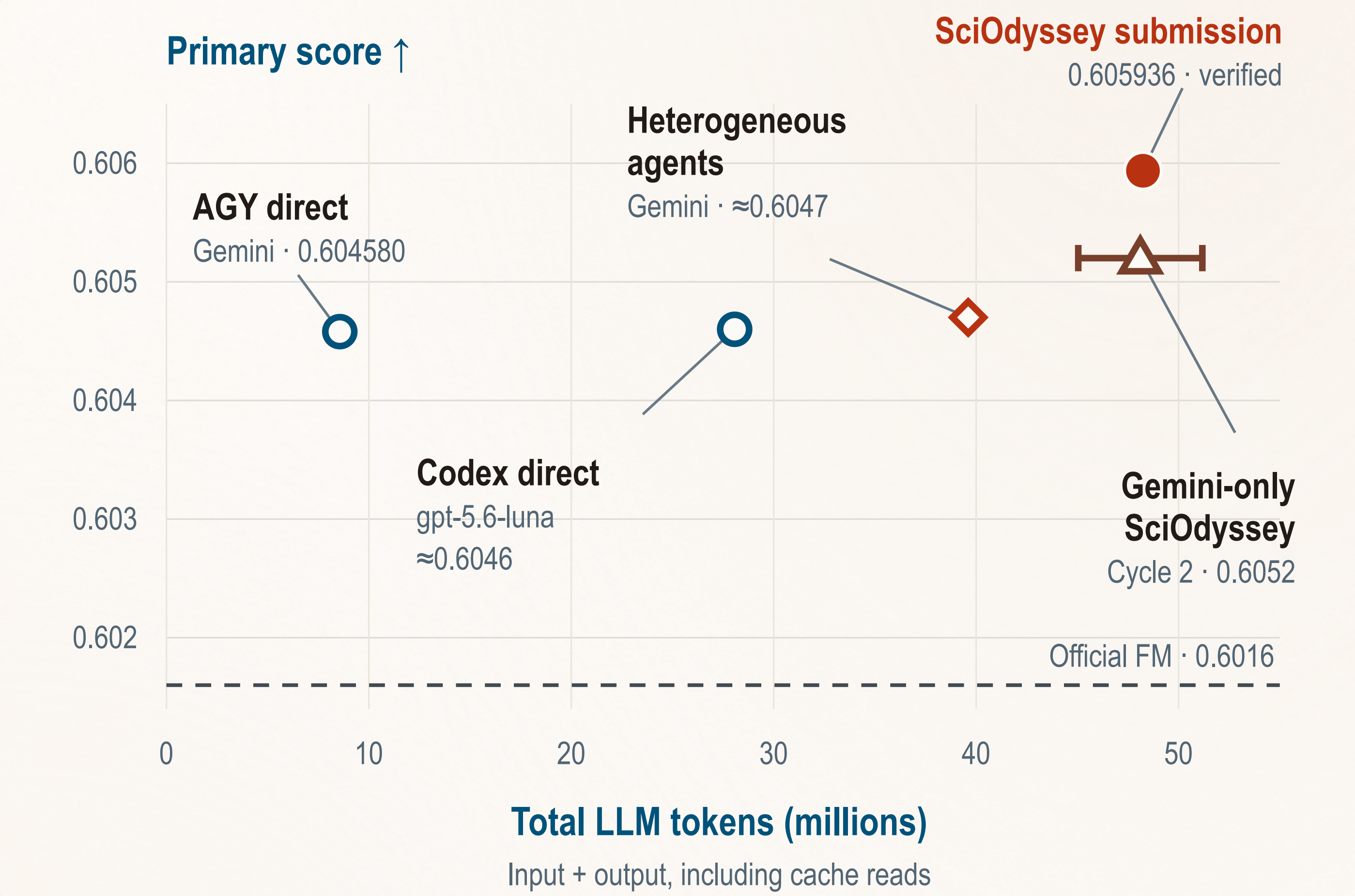
One agent. Any ML task. End to end.

task.md What to solve Objectives · constraints · evaluation	PERSONAL.md How you want it solved Preferences · research style · priorities
Define the task →	Delegate research → Review evidence
EXAMPLE INPUT	
Use the research-agent skill for task.md. Follow PERSONAL.md. Run autonomously. Let me review the results and evidence.	

05 / EVALUATION

Research outcomes, in context.

KuaiRand-Pure · public validation · one recorded run per condition



* Comparative exploration across distinct agent setups under KuaiRand-Pure benchmark constraints.

Verified submission · public validation

Metric	Official FM	SciOdyssey	Δ
Primary	0.6016000	0.6059363	+0.0043363
GAUC	0.6674000	0.6728421	+0.0054421
nDCG@5	0.5357000	0.5390304	+0.0033304

Primary = mean(GAUC, nDCG@5). Final recipe: 46 categorical features, eight-seed FM ensemble, NumPy / CPU.

4 autonomous cycles | 13 named experiments | 0 manual scientific interventions*

7 Full evaluations · 0 GPU-hours · 170,588 prediction rows passed the unchanged Starter Kit alignment checker.

* 100% autonomous execution throughout the retained submission record.

06 / INSIGHTS

Observed behavior, not just a score.

Recovery NumPy float32 values broke serialization after evaluation. The Scientist repaired the writer and retained the evidence.	Audit Reviews caught leakage and metric issues early. META audits empirical claims against artifacts before promoting State.	Insight synthesis Heterogeneous Scientists turn diverse search styles into a broader exploration prior; parallel search produces stronger candidates.
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BETTER PROCESSES.
A WIDER RESEARCH WORLD.